Hybrid Photogrammetry Structure-from-Motion Systems for Scene Measurement and Analysis

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1. Introduction

The reconstruction of scenes from multiple images or video streams has become an essential component in many modern applications. Many fields, e.g. Robotics, surveillance, and virtual reality, employ systems to reconstruct a three-dimensional (3D) representation of an environment in order to accomplish their specific goals. There are a number of systems [1, 2, 3] capable of reconstructing scenes from thousands of images. Increasing the accuracy of these systems are still areas of ongoing research.

In the field of metrology, photogrammetry is often used to obtain a 3D representation of an object or scene [REFERENCE]. Often, targets (individual markers) are placed on the object and are used to identify points of interest on the object. The targets are usually retro-reflective and can be coded to easily identify one target from another. Non-coded targets are employed as well and are identified in relation to a coded target. The choice and placement of targets depend on the desired measurement. Photographs of the object are taken from various camera positions ensuring that the same target appears in at least two images. The target center for each target is found in the images where it appears and triangulation is used to determine to find the 3D coordinate of each target. The resulting point cloud can later be processed to obtain a 3D polygonal mesh or other representation [3].

In computer vision, the technique used to obtain a 3D model of an object is known as structure from motion (SfM). While SfM has developed into a subfield of computer vision, the inception of this field stems from photogrammetry. The fundamental mathematics used in SfM are also present in photogrammetry and many algorithms overlap the two fields. SfM involves a number of methods common in the computer vision and photogrammetry literature, such as feature tracking (target tracking), frame decimation, self-calibration, camera pose estimation (known in photogrammetry as camera calibration), triangulation, and bundle adjustment [4]. Despite the similiarities between the fields, the goals of the fields have come to diverge. Generally, photogrammetry is focused on obtaining surface points on the object that have very low covariances. SfM typically favors dense reconstructions so that more knowledge of the scene can be obtained. Visually accurate results are obtained with SfM that are not sufficient for metrology applications. However, there is recent work in SfM to obtain more accurate results [5].

Given the photogrammetry and SfM advances, an application harnessing the symbiotic relationship between these fields is now possible. The system uses photogrammetric information to enhance the accuracy of SfM. SfM provides a more complete reconstruction, where photogrammetry is used to correct camera parameters or scene points that have higher uncertainty. This procedure is less expensive, maintains the required metrological accuracy, and provides a dense reconstruction. In addition, an analysis of the effects from various camera parameters, such as focal length and depth, on a reconstruction was performed to determine the correct setup for optimal results. An overview of related technologies is provided in Section 2. The components of the system are presented in Section 3. Experimental results are discussed in Section 4, conclusions in Section 6, and future work in Section 7.

2. Background

Multi-view reconstruction attempts to obtain a 3D representation of an underlying scene from a collection of images taken from various different camera viewpoints. Rays from each camera center through its image plane define the 3D intersection point for corresponding pixels between images. The output of scene reconstruction is characteristically a

point cloud, though post-processing can be applied to obtain a polygonal mesh or other volume representation. A set of corresponding pixels, known as feature tracks, can be computed using sparse or dense tracking algorithms. This is the most important process in scene reconstruction, as errors in this stage will affect all subsequent stages.

The following sequential stages are necessary for performing multi-view reconstruction. However, there exist many different algorithms for this purpose and these are the most common steps. Typically, feature matches (between consecutive pairwise views) and tracks (concatenating across multiple views) are first obtained. Such tracking can be sparse [6, 7] or dense [8] and consists of computing and linking the pixel coordinates in all images for each scene point, whenever it is visible. Frame decimation [9] should be applied at this point, particularly for sequential image sets, to filter out images with very small or very large baselines. A baseline is the relative separation of the images in world space. Small baselines lead to bad numerical conditioning in the pose and structure estimation. Large baselines introduce problems in feature tracking. Next, camera intrinsics are estimated through a process called self-calibration [4]. In most cases, this information is already known. Also, the epipolar geometry can be estimated from pairs of views or triplets. The epipolar geometry encapsulates the intrinsic projective geometry between groups of views and is used in the process of determining the camera pose (position and orientation). Only relative positions and orientations can be obtained between the pairs or triplets of views. Once the camera parameters are estimated, computation of the scene structure can be achieved through triangulation methods, such as "linear triangulation" [4]. In this method, the 3D position of a scene point, given a set of cameras and pixel feature track positions corresponding to the point, is computed as the best-fit intersection position in space for the set of rays from each camera center and through the feature track positions. Finally, since errors in all of the above steps influence the accuracy of the computed structure, bundle adjustment is performed to optimize all or a number of the different camera and structure parameters [10, 11]. Usually bundle adjustment employs the Levenburg-Marquardt algorithm to minimize 'reprojection error' of all computed structure, however the algorithm may converge to a local minimum and not achieve the globally optimal result.

As for specific scene reconstruction algorithms, many of which are based on all or a combination of the steps outlined above, there are numerous successful algorithms currently in the literature. Examples include: Akbarzadeh et al. [12] who introduced a method for dense reconstruction of urban areas from a video stream and Snavely et al. [1] who presented a system for interactively browsing and exploring large unstructured collections of photographs of a scene. Snavely et al.'s system used an image-based modeling front end that automatically computes the viewpoint of each photograph as well as a sparse 3D model of the scene. The most recent system that incorporates many modern reconstruction algorithms is Wu et al.'s VisualSFM [2].

There has been recent work in the computer vision literature that focuses on how different parameters in a reconstruction affect the final output. Beder and Steffen [13] introduced an uncertainty metric for a single scene point computed using linear triangulation. They proved that the optimal answer for an N-view triangulation lies within this 3D uncertainty ellipse. Knoblauch et al. introduces an metric for determining the source of error for a given feature match [14]. Their method did not rely upon any apriori scene knowledge. More recently, Recker et. al [15] introduced a scalar field visualization system based upon an angular error metric for understanding the sensitivity of a 3D reconstructed point to various parameters in multi-view reconstruction. This work investigates the advantages that SfM and photogrammetry can gain from each other as well as understanding how different parameters affect the quality of a reconstruction.

3. Methodology

The proposed system utilizes photogrammetry to enhance the accuracy of SfM, which provides a more complete scene reconstruction. In addition, SfM is useful towards improved underlying scene analysis in photogrammetry. The hybrid system diagram is presented in Fig 1.

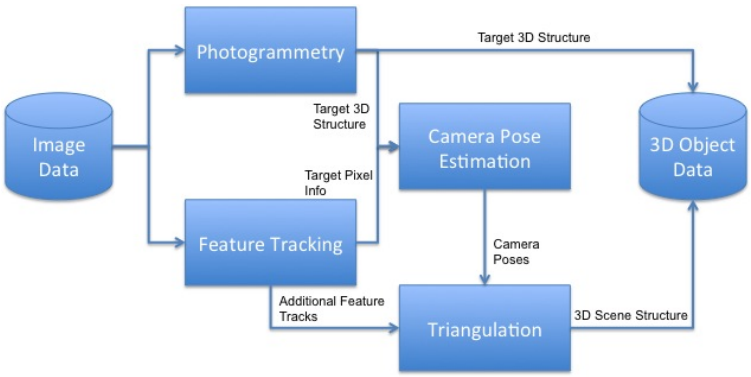


Figure 1: This figure demonstrates the processes used in the presented system. The algorithm begins by identifying the targets in the image sequences. In addition, the images are processed using an automatic feature tracking algorithm, in this case SIFT [6]. Camera calibration and triangulation of the targets is performed. The camera parameters are passed to the scene triangulation stage and are used to triangulate the remaining scene points obtained from the feature tracking stage. Bundle adjustment is performed to optimize the remaining structure. The point cloud is stored.

First, images of a targeted asset are provided to a photogrammetry system and the 3D target positions are computed. Additionally, the images are also processed for additional feature tracks. This implementation used SIFT [6] but additional techniques such as SURF [7] and DAISY [8] could be used. Some photogrammetry systems also provide the camera intrinsics and pose. In the event this information is not provided, the camera pose for all images is estimated using the 3D target positions and 2D projections by solving the perspective N-point problem (PnP). This was accomplished using the Efficient Perspective N-Point (EPnP) algorithm [16]. Finally, any additional feature tracks are triangulated, using Recker et al's [5] triangulation scheme and bundle adjustment [17] is performed to optimize the structure and computed camera pose. The point cloud is stored and can be post-processed to obtain another representation.

4. Results

The proposed hybrid algorithm was tested extensively for its accuracy and general behavior, on both real and synthetic data. All implementation was done using C++ on a MacBook Pro with an *Intel Core i7* processor at 2.66 GHz with 4GB of RAM, running Mac OS X Mavericks 10.9.1.

4.1. Synthetic Tests

A variety of synthetic scenes were used to verify elements of the hybrid techique. Computation of the scene's 3D structure is accomplished using linear triangulation, whose accuracy depends upon the previously computed camera projection matrix and feature track position. This set of tests examined the behavior of these parameters on the resulting 3D structure.

Three different error metrics were recorded for the experiments: reprojection error, rotational error, and positional error. The formulas are shown in Eqs. 1, 2, and 3.

$$f_{reprojectionerror}(X, P) = ||PX - x|| \tag{1}$$

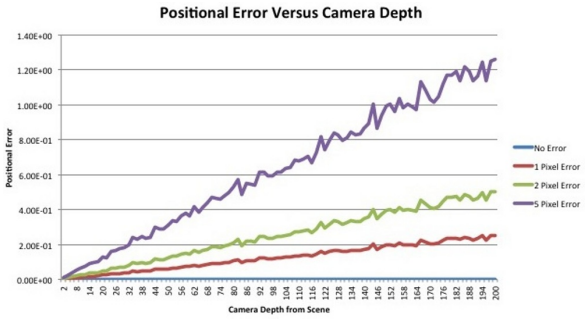
$$f_{rotationalerror}(R_i, R_j) = \frac{||R_i - R_j||}{||R_j||} \tag{2}$$

$$f_{positionalerror}(X_i, X_j) = ||X_i - X_j|| \tag{3}$$

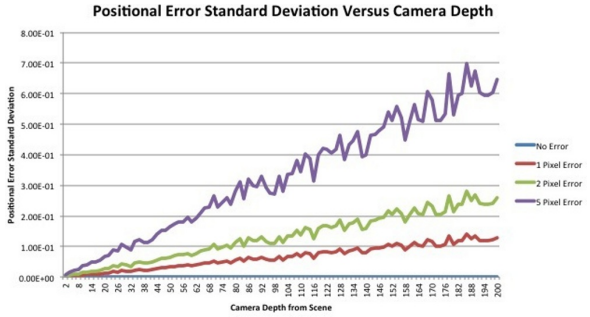
Here X is a 3D scene point, P is a 3×4 camera projection matrix, and x is a 2D feature track pixel location corresponding to X. R_i and R_j are quaternions that encapsulate a 3D rotation. Also, X_i and X_j are 3D positions.

4.1.1. Analysis of Feature Tracking Noise and Camera Depth

The first experiment in this set examined the behavior that camera depth has on the accuracy of a 3D reconstruction. A ground-truth camera was positioned at increasing distances, in units of 2 along the Z-axis, of a $1 \times 1 \times 1$ box with 100 ground truth 3D positions, X_i, on the surface of the box. The ground truth positions were projected into each camera position, denoted x_i. At each distance, image noise was introduced using a zero-mean Gaussian distribution with increasing standard deviation, σ, from 0 to 5 pixels. The original distance, d_i, from the camera center and each ground truth point was computed. Then using the noisy 2D projection, x_i', a ray, with unit direction, from the camera center through x_i' was computed. A 3D point along that ray was computed using d, denoted X_i'. The positional error between X_i and X_i' was computed for all 100 scene points. The results were averaged and the standard deviation was computed, and displayed in Fig 2.



(a) Positional error vs depth from camera



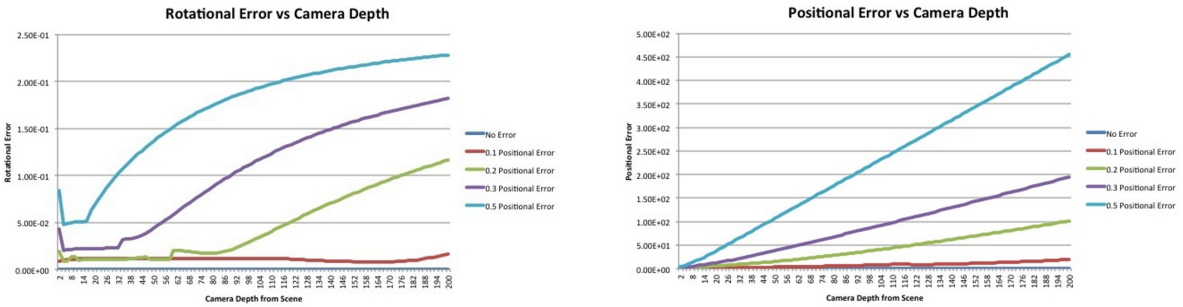
(b) Positional error standard deviation vs depth from camera

Figure 2: This figure demonstrates the results of increasing the camera depth and observing the effect of 3D structure computation in the presence of image noise. A ground-truth camera was positioned at increasing distancesof a $1 \times 1 \times 1$ box with 100 ground truth 3D positions. At each distance, image noise was introduced using a zero-mean Gaussian distribution with increasing standard deviation from 0 to 5 pixels. The original distance from the camera center and each ground truth point was computed. Using the noisy 2D projection, a ray, with unit direction, from the camera center is computed. A 3D point along that ray was computed using the original distance. The positional error between the two points is computed. The results were averaged, as shown in (a) and the standard deviation was computed, as shown in (b). It is easily observed that as the camera is moved away from the scene any inaccuracies in feature tracking manifest in larger distances from the ground-truth point.

The effects of depth can easily be seen from Fig 2. As the camera is moved away from the scene, moving the feature track location on the image plane even small amounts manifests in 3D computed positions that are large distances away from the original data. This demonstrates two important principles to consider when reconstructing objects from images. First, the object should be reasonably close to the camera. This allows the tracking to be slightly inaccurate without affecting the 3D computed position too much. Now, as the distance of the cameras from the object increases, the computed 3D position deviates further from the true position. This observation leads to the second principle: if the object is far from the cameras the tracking should be as accurate as possible. This will lead to the most accurate 3D position to be computed.

4.1.2. Camera Pose Estimation with Positional Noise and Camera Depth

The second experiment used the same setup as the previous experiment. However, in this experiment the ground-truth 2D projections were kept fixed. 3D positional noise was introduced to the box structure, again sampling from a Normal distribution with zero mean, and σ increasing from 0 to 1. The experiment ensured that all 3D points were positioned in front of the cameras. For each depth, the camera pose was solved and reprojection error, rotational error, and positional error were recorded. For rotational and positional error, the ground truth camera was used as reference. The results of this test are shown in Fig 3.



(a) Rotational error vs pose

(b) Positional error vs pose

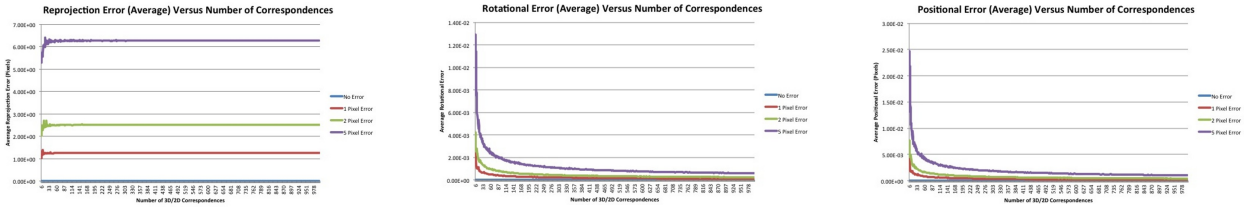
Figure 3: This figure demonstrates the results of solving for camera pose when introducing positional noise into ground truth 3D points and solving for cameras that increase in depth away from the scene. The scene setup was the same as in Fig 2. Here the 3D positions were varied according to a Normal distribution with zero mean and increasing σ from 0 to 1 in steps of 0.1. The rotational error and positional error were recorded with respect to the original ground-truth camera data. It is easily observed that as the scene gets further away the accuracy of the computed pose decreases.

4.2. 3D Model Tests

Two synthetic datasets were used, *Bradley* and *ch47* to again analyze camera pose estimation. The objects were rendered using known camera parameters and ground-truth feature tracks were generated from the 3D object data.

4.2.1. Camera Pose Estimation in the Presence of Image Noise

The first experiment in this set determined the accuracy of the computed camera parameters when introducing image noise. Image noise was sampled from a Gaussian distribution with zero mean and standard deviation increasing from 0 to 10 pixels. The camera parameters were estimated and reprojection error, rotational error, and positional error were recorded for each camera. *Bradley* contained 1234 cameras. The error metrics were averaged and standard deviation was computed. Additional experiments were conducted that varied the number of ground-truth 3D points that were used. The number of points varied from 5 to 1000. The results of this first experiment are presented in Fig 4.



(a) Average reprojection error

(b) Average rotational error

(c) Average positional error

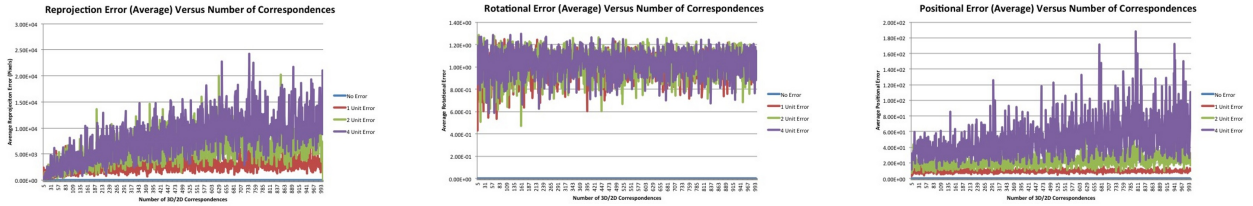
Figure 4: This figure demonstrates the results of computing camera pose in the presence of image noise. The results from *bradley* are shown. Average reprojection error is shown in part (a), average rotational error in part (b), and average positional error in part (c).

From Fig 4, for low numbers of 3D/2D correspondences there is a larger uncertainty in the computed camera pose. When the number of correspondences is increased, the error tends to stablize for all error metrics that were computed and across the different noise levels.

4.2.2. Camera Pose Estimation in the Presence of 3D Point Error

The second experiment determined the accuracy of the computed camera parameters when introducing 3D positional noise. Positional noise was sampled from a Gaussian distribution with zero mean and standard deviation increasing from 0 to 5 world space unit. The camera parameters were estimated and reprojection error, rotational error, and positional error were recorded for each camera. The error metrics were averaged and standard deviation was computed.

Additional experiments were conducted that varied the number of ground truth 3D points that were used. The number of points varied from 5 to 1000. The results of this second experiment are presented in Fig 5.



(a) Average Reprojection Error (b) Average Rotational Error (c) Average Positional Error

Figure 5: This figure demonstrates the results of computing camera pose in the presence of 3D positional noise. The results from *bradley* are shown . Average reprojection error is shown in part (a), average rotational error in part (b), and average positional error in part (c).

From Fig 5, the overall sensitivity to the 3D structure is greatly increased when compared to introducing 2D image noise. For large movement in the 3D point the camera pose estimate becomes less accurate. When compared to the original camera data that was used to generate the 2D projections, the reprojection error is much improved.

4.2.3. Camera Pose with Different Spatial Distributions of Points

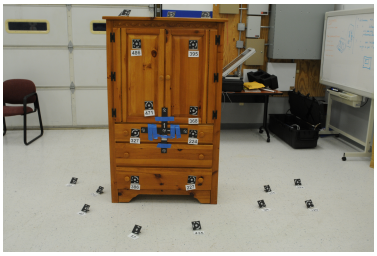
The final experiment determined the accuracy of the computed camera pose when solving the pose using different spatial distributions. The first distribution of points used was a small cluster of points and the second was a random distribution. The camera parameters were estimated the same error metrics in the previous tests were used. The number of points used was 50. The results of this experiment are presented in Fig 6.

Figure 6: This figure demonstrates the results of computing camera pose using different spatial distributions of 3D scene points. For different spatial distributions, the accuracy of the camera pose is unchanged.

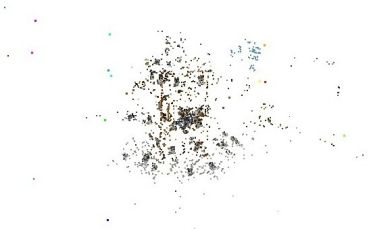
The results of this test indicate that the spatial distribution of the 3D scene points have no effect on the estimation of the camera pose. Provided accurate tracking and 3D points are provided, EPnP solves for the camera pose no matter the distribution of points. In the presence of error, the reprojection error is minimized which could result in a large change in camera parameters from ground truth measurements.

4.3. Real Tests

To verify the hybrid technique, a real dataset was run on the system comparing to VisualSFM [2], a state-of-the-art SfM system. The first dataset contained 14 images of an armoire containing 37 targets were taken. The AICON DPA Pro ® system solved for the targets 3D positions. All images had the same camera intrinsics, provided by the AICON system. The camera intrinsics and target positions were entered into HawkEye ®, a SURVICE SfM library. HawkEye computed feature tracks using SIFT [6] and triangulated the results. Fig 7 demonstrates the results of the hybrid technique against VisualSFM. The hybrid results have 6659 total computed scene points with an average reprojection error of 0.4376 pixels compared to 3073 points with VisualSFM and an associated average reprojection error of 6.1426 pixels.



(a) Reference image



(b) Hybrid result



(c) VisualSfM result

Figure 7: This figure contains the results of the *armoire* dataset. A reference image is shown in (a). (b) shows the hybrid point cloud with 6659 points and average reprojection error of 0.4376 pixels. (c) shows the results of VisualSFM, which has a total of 3073 points and average reprojection error of 6.1426 pixels.

5. Practical Guidelines

The experiments in the results section demonstrate the effect that certain parameters have on the accuracy of a 3D reconstruction. Camera depth from the scene is an important factor to consider. It has been shown that the closer the camera is to the scene, the less effect a tracking error has on the final reconstruction. When the camera is far from the scene, the effects of tracking errors are amplified. However, it is also shown that having a camera that is too close to the scene can be problematic for rotation estimation and that there is a range of good distance values that result is good camera rotation estimations.

Camera depth is not the only factor to consider. In addition, focal length and image resolution also affect the accuracy of a reconstruction. Unfortunately, these three factors are not independent of one another. Focal length specifies the distance between the image plane and the camera center of projection. Increasing the focal length, decreases the area that is encapsulated by a single pixel. Large focal lengths result in a narrow field of view and thus less of the scene can be captured by a single image. To capture more of the scene while minimizing the amount of area a single pixel covers, image resolution can be increased. However, increasing the focal length too much can nullify the increased image resolution benefits. On a practical level, having a pixel cover too much area can lead to inaccuracies in target location. If the pixel covers a significant portion of the target, then there will only be a few pixels in which the target is present. This will lead to poor target tracking and thus a poor reconstruction in general.

Another factor that can affect the quality of a reconstruction is lighting. For automated feature tracking techniques, for example SIFT [6], changes in lighting can cause mis-matches or for the same points to not be identified. Ideally, lighting should approximate a constant global illumination from all angles (i.e hemisphere) for best results. If uniform lighting is not possible then an overhead lightsource can work, however care must be taken to avoid drastic changes with auto white balancing in the images. For example, if an image of the light source is present in the image, the resulting image is often much darker than it should be. This can lead to feature tracking errors.

6. Conclusion

In conclusion, this paper presents a hybrid technique that combines photogrammetry with structure-from-motion (SfM) techniques. The system uses photogrammetric information to enhance the accuracy of SfM. SfM provides a more complete reconstruction, where photogrammetry is used to correct camera parameters or scene points that have higher uncertainty. This procedure is less expensive, maintains the required metrological accuracy, and provides a dense reconstruction. Results of the hybrid system were demonstrated on both real and synthetic data, where it has shown to provide accurate and dense reconstructions.

7. Future Work

The development of hybrid systems for 3D reconstruction helps improve both fields of photogrammetry and structure-from-motion. The continued exploration of combining both fields can lead to improvment of current algorithms and the development of new ones. Investigation into photogrammetry assisted volume based reconstruction is interesting and important for future applications. Additional visualization techniques are needed to show reconstruction accuracy

and where additional photogrammetry can be performed to improve the overall reconstruction accuracy.

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